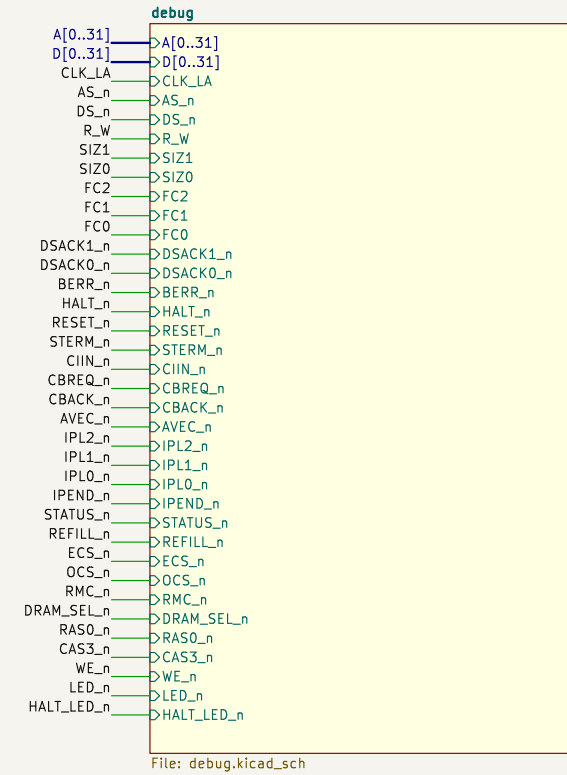
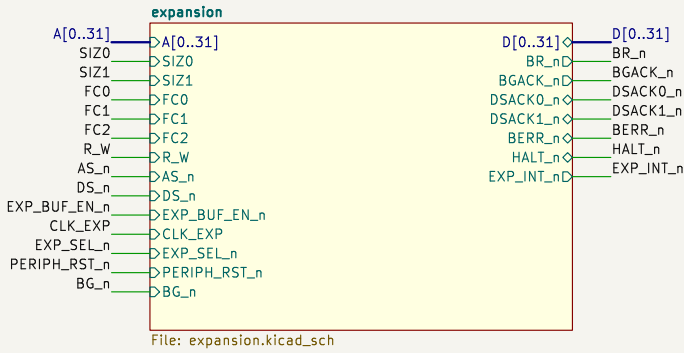
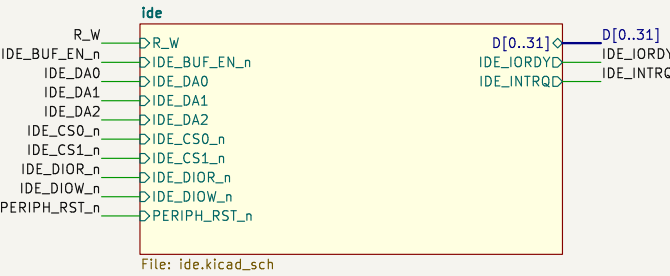
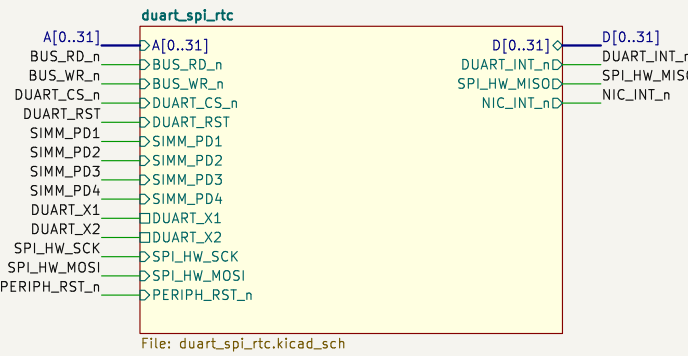
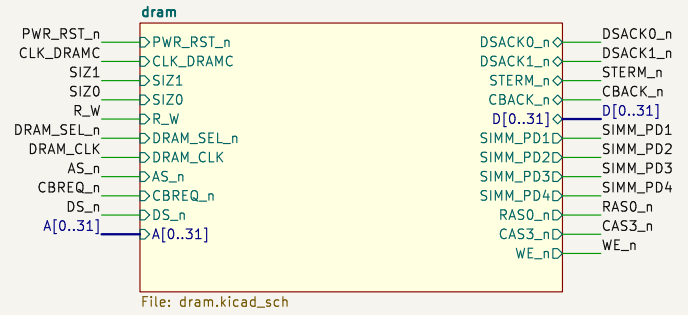
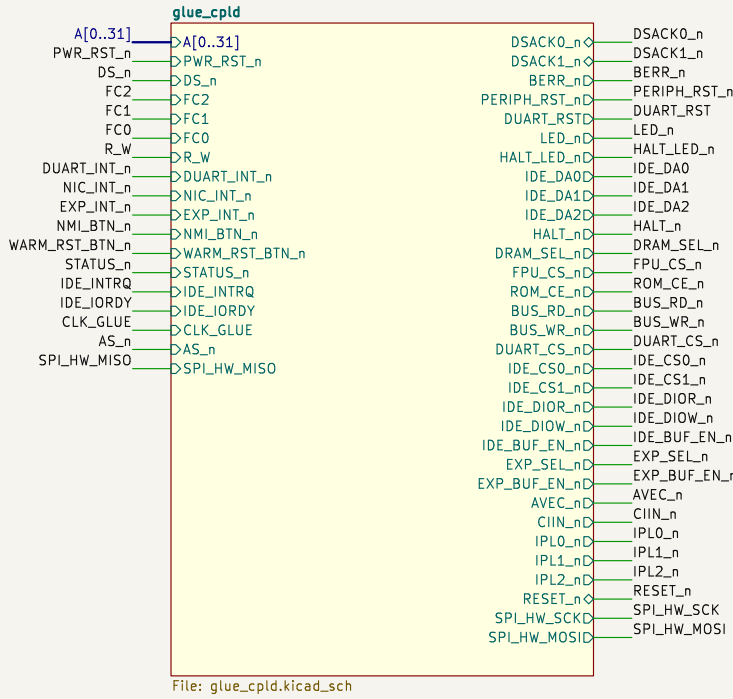
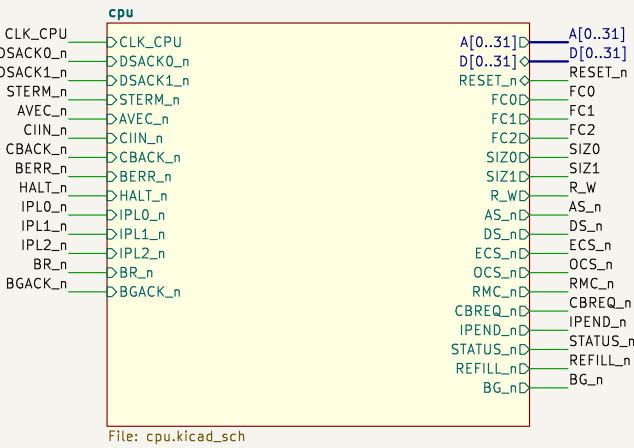
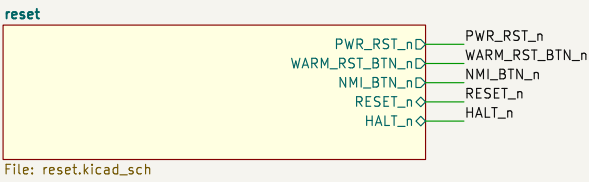
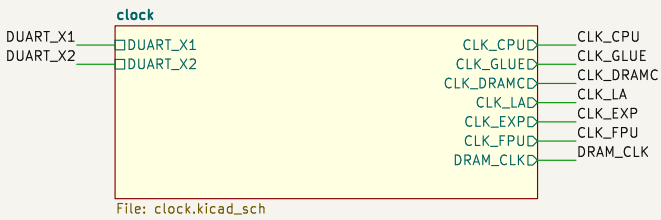
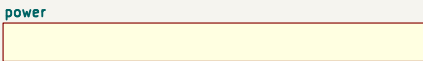
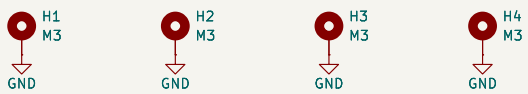


68030 Linux SBC – rev A draft (KiCad 9). Sheets in page order: 2 power, 3 clock, 4 reset, 5 cpu, 6 fpu, 7 glue_cpld, 8 dram, 9 rom, 10 duart_spi_rtc, 11 ide, 12 expansion, 13 debug. All sheets drawn.



Mounting holes (plated, to GND)



Attribution: lib/m68030-sbc.pretty/PGA128_13x13_MC68030 and SIMM-72_Mackerel, and the MC68030 / MC68882 pin maps, are derived from Mackerel-68k by Colin Maykish (MIT license, see lib/LICENSE-mackerel-68k.txt). Pin numbers re-verified against the MC68030 User's Manual sect. 14 and MC68882 BR509 – see NOTES.md.

Drawn: power, clock, reset, cpu, fpu, glue_cpld, dram, rom, duart_spi_rtc, ide, expansion, debug. No test points.
PGA-128/SIMM-72 footprints + pin maps: Mackerel-68k (MIT, C. Maykish)
68030 Linux SBC – hierarchical top level. Design spec: ../spec.md

Matthew Bettcher – 68030 Linux SBC

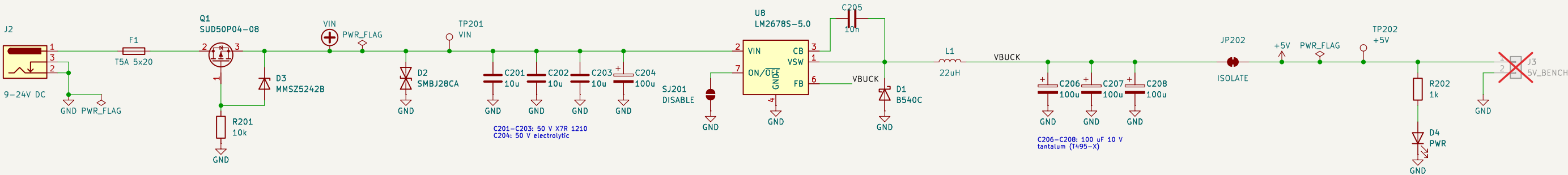
Sheet: /
File: m68030-sbc.kicad_sch

Title: 68030 Linux SBC – top level

Size: A2 Date: 2026-09-26 Rev: A (draft)
KiCad E.D.A. 9.0.2 Id: 1/13

Input: 9–24 V DC barrel jack (centre +), T5A fuse, P–FET reverse–polarity protection, TVS

5 V buck: LM2678S–5.0 (fixed 5 V, 260 kHz, 5 A switch)

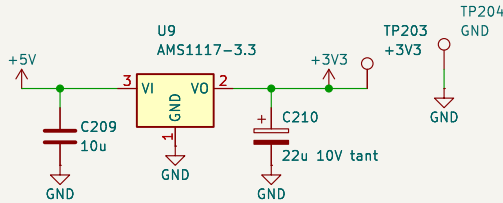


Q1: body diode conducts first, then $V_{gs} = -(VIN)$ turns the channel on; reversed input leaves Q1 off. D3 clamps $|V_{gs}|$ to 12 V (Q1 V_{gs} max ± 20 V). D2 SMBJ28CA: bidirectional TVS, 28 V standoff; clamp at rated lpp (~ 45 V) is at the LM2678 45 V abs–max [EST] – fuse F1 clears sustained faults.

C201–C204: Irms rating must exceed $I_{load}/2 = 2$ A (SNVS029L step 5). Deviation from Table 7–7 (tantalum/aluminium): 3x 10 uF 50 V X7R 1210 + 1x 100 uF 50 V low–ESR electrolytic for bulk/damping on hot–plug.

L1: Fig 7–3 nomograph (5 V, 24 V, 4 A) $\rightarrow$ L41 = 22 uH; SRP1770TA–220M: DCR 26.5 mOhm, Isat 18 A. C206–C208: Table 7–5 $\rightarrow$ 3x AVX/KEMET 100 uF 10 V ‘C4’ tantalum (no all–ceramic output). D1: Schottky, VR $\geq 1.3 \times 24$ V, IF ≥ 5 A (Table 7–4 class) $\rightarrow$ B540C (40 V, 5 A). C205 (CB): 0.01 uF per step 7 / sect. 7.1.6 (pin table says 100 nF – see NOTES.md). Output: 4.85–5.15 V over line/load/temp (SNVS029L electrical table). SJ201: ON/OFF (pin 7) left open = enabled (abs max 6 V, never tie to VIN); bridge to disable. JP202: cut to isolate the buck and feed +5V from J3 (bench supply). J3 DNP.

3.3 V LDO for DS3234 RTC / ENC28J60 / logic–level parts (spec 6.1)



C210: AMS1117 datasheet ‘Stability’: 22 uF solid tantalum on the output covers all conditions. C209: 10 uF input per the datasheet application circuit.

Mounting holes H1–H4 (GND) are on the top–level sheet.

Load budget and rail list: spec.md sect. 6.1		
Input: fuse $\rightarrow$ P–FET reverse–polarity switch $\rightarrow$ TVS on VIN		
LM2678 values: TI SNVS029L sect. 7.2.1.2, Vin(max) 24 V, Iload 4 A (spec 6.2)		
Matthew Bettcher – 68030 Linux SBC		
Sheet: /power/		
File: power.kicad_sch		
Title: Power: 9–24 V input, LM2678–5.0 buck, 3.3 V LDO		
Size: A3	Date: 2026–09–26	Rev: A (draft)
KiCad E.D.A. 9.0.2		Id: 2/13

CLKB_CPU → R301 → CLK_CPU

CLKB_FPU → R302 → CLK_FPU_CPU

CLKB_GLUE → R303 → CLK_GLUE

CLKB_DRAMC → R304 → CLK_DRAMC

CLKB_LA → R305 → CLK_LA

CLKB_EXP → R306 → CLK_EXP

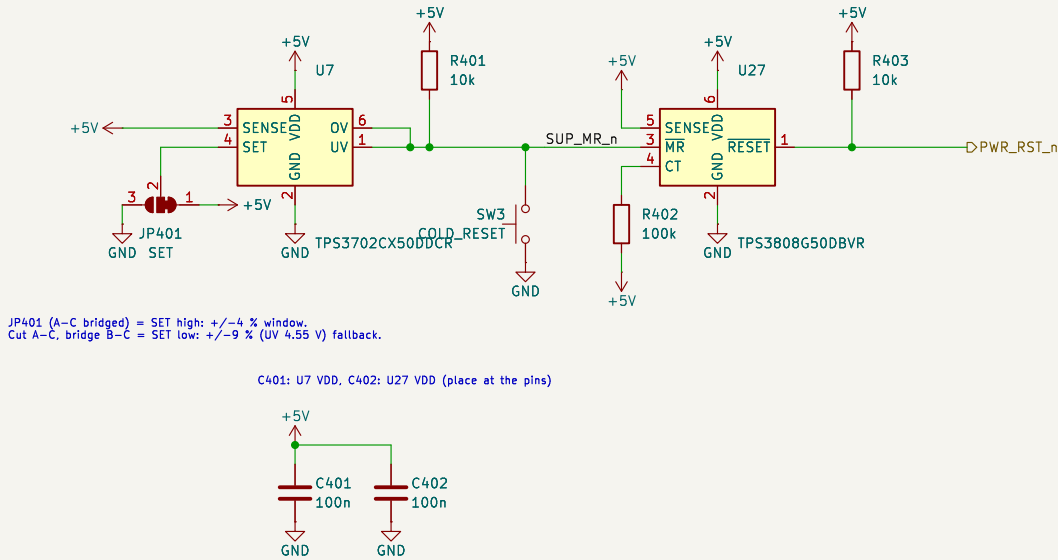
CLK_CPU → U1 CLK (E1); CLKB_GLUE → U3 pin 83 (GCLK1);
 CLKB_DRAMC → U4 pin 2 (GCLK2); CLKB_LA → LA3; CLKB_EXP → J10

The schematic diagram shows the X2 module, a yellow square component with four pins. Pin 1 is connected to a +5V supply. Pin 4 is also connected to a +5V supply. Pin 7 is connected to a GND symbol. Pin 8 is connected to a 33R resistor, which is then connected to the DRAM_CLK signal. The module is labeled 'X2' and '50.000MHz'. Below the module, there is a decoupling circuit for the +5V supply, consisting of two capacitors, C305 (100n) and C306 (10u), connected in parallel to GND.

307/C308: 22 pF [EST] – size to the crystal's CL (spec 7.1).

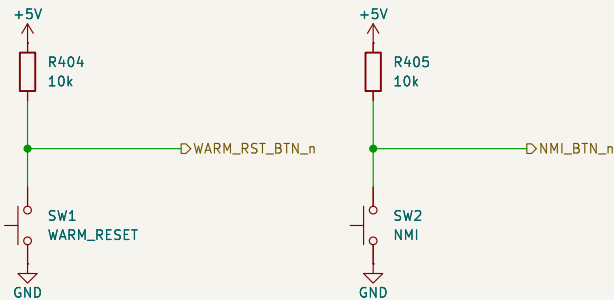
Place Y1 next to U6 pins 36/37 (DUART sheet)
 68030 needs a stable, continuous clock before VCC is valid (UM 7.8)
 spec 7.1: socketed X1 (25 -> 33.333 MHz), 74ACT244 fan-out, 33R series [EST]
Matthew Bettcher — 68030 Linux SBC
 Sheet: /clock/
 File: clock.kicad_sch

Power-on / brown-out supervisor

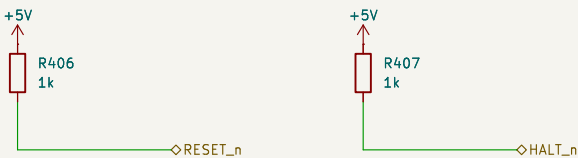


Operation: U7 compares +5V against a 4.757–4.843 V (UV) / 5.153–5.247 V (OV) window; its open-drain outputs, SW3 and R401 form SUP_MR_n into U27 MR. U27 holds PWR_RST_n low while MR is low or +5V < 4.65 V, then releases it ~300 ms later (CT = 100k to VDD). LM2678–5.0 output: 4.85–5.15 V (TI SNVS029L).

Buttons (debounced in the glue CPLD, spec 3.5 / 7.2)



CPU RESET_n / HALT_n open-drain pull-ups (spec 8.1)



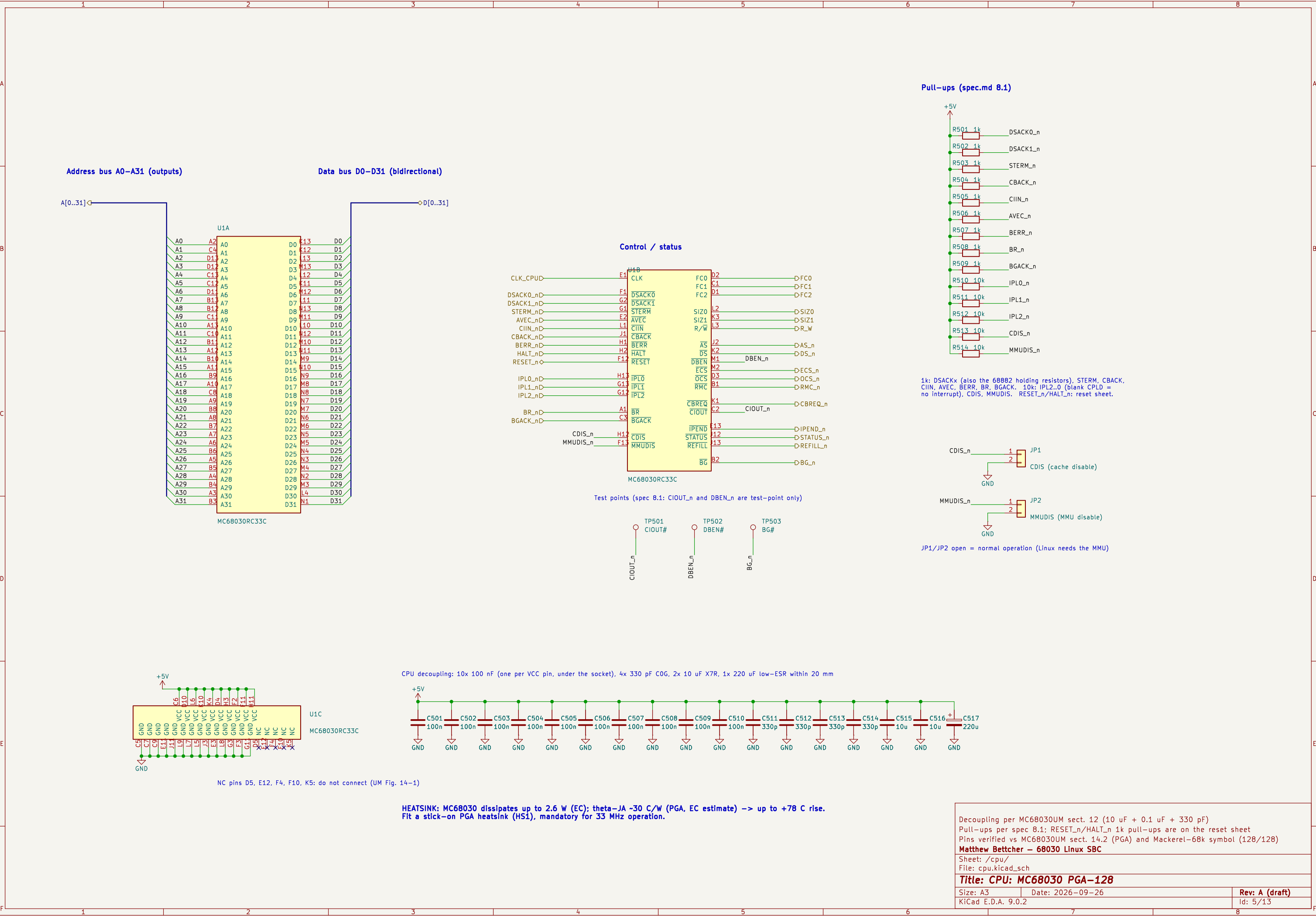
RESET_n: driven by U3 (power-on / warm reset), by the CPU RESET instruction, read by U3; also FPU RESET and LA3. HALT_n: 68030 Input only, driven by U3 with RESET. 1k gives ~5 mA sink at 5 V (CPU IOL 10.7 mA, EC).

PWR_RST_n -> U3 pin 1 and U4 pin 1 (GCLR); spec 7.2
RESET_n / HALT_n are open-drain nets; 1k pull-ups here (spec 8.1)
TPS3702CX50 (SBVS251A Tbl 10–2) window + TPS3808G50 (SBVS050N) delay
Matthew Bettcher – 68030 Linux SBC

Sheet: /reset/
File: reset.kicad_sch

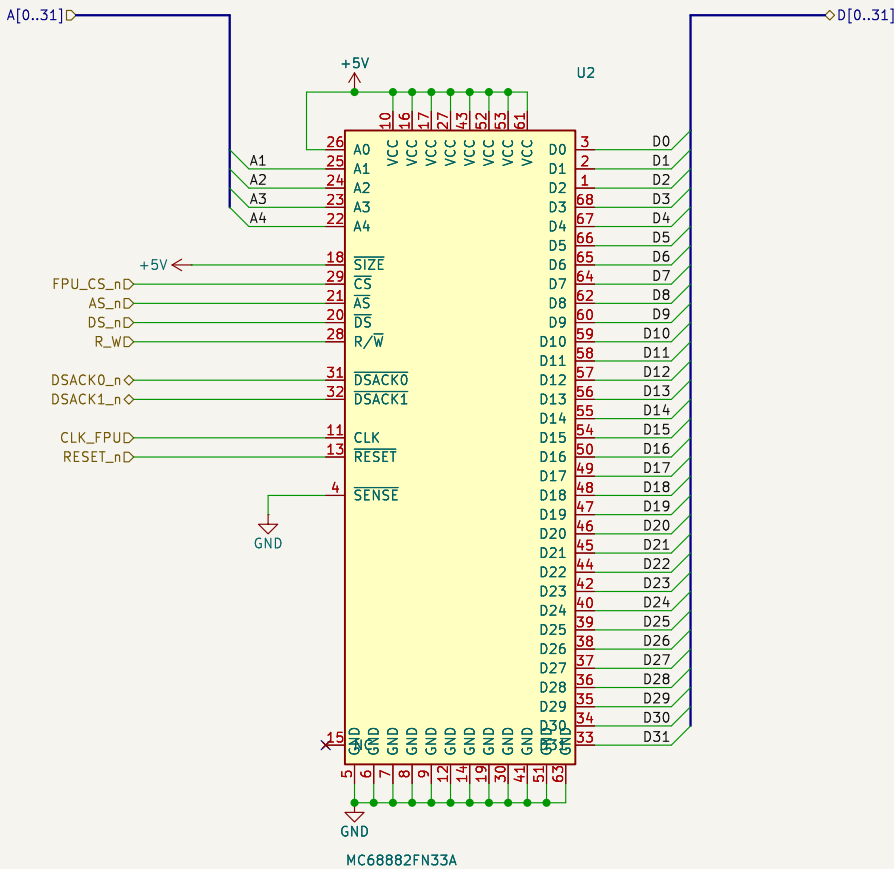
Title: Reset

Size: A3	Date: 2026–09–26	Rev: A (draft)
KiCad E.D.A. 9.0.2		Id: 4/13

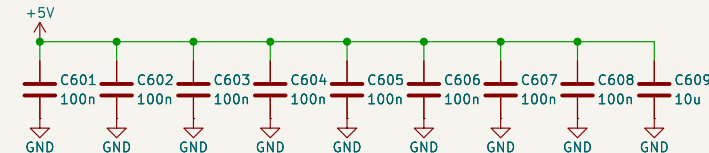


Notes (spec.md 8.2):

- DSACK1_n/DSACK0_n connect directly to the CPU DSACK nets (three-state, actively negated then released; the 1k pull-ups on the CPU sheet are the holding resistors, UM 9.8).
- SENSE tied to GND (UM 9.11; the optional presence-detect circuit is not used).
- CLK_FPU comes from JP3 on the clock sheet (1-2 = CPU clock, 2-3 = optional X3).
- RESET shares the CPU RESET_n net.



FPU decoupling: 8x 100 nF (one per VCC pin: 10, 16, 17, 27, 43, 52, 53, 61) + 10 uF



FPU must be rated $\geq$ its clock (FN25A/FN33A), else use X3 + JP3
32-bit port: A0 and SIZE tied to VCC (68881/2 UM / BR509)
PLCC-68 pins verified vs BR509 Rev.3 p.23 and Mackerel-68k (68/68)

Matthew Bettcher - 68030 Linux SBC

Sheet: /fpu/
File: fpu.kicad_sch

Title: FPU: MC68882 PLCC-68

Size: A3
KiCad E.D.A. 9.0.2

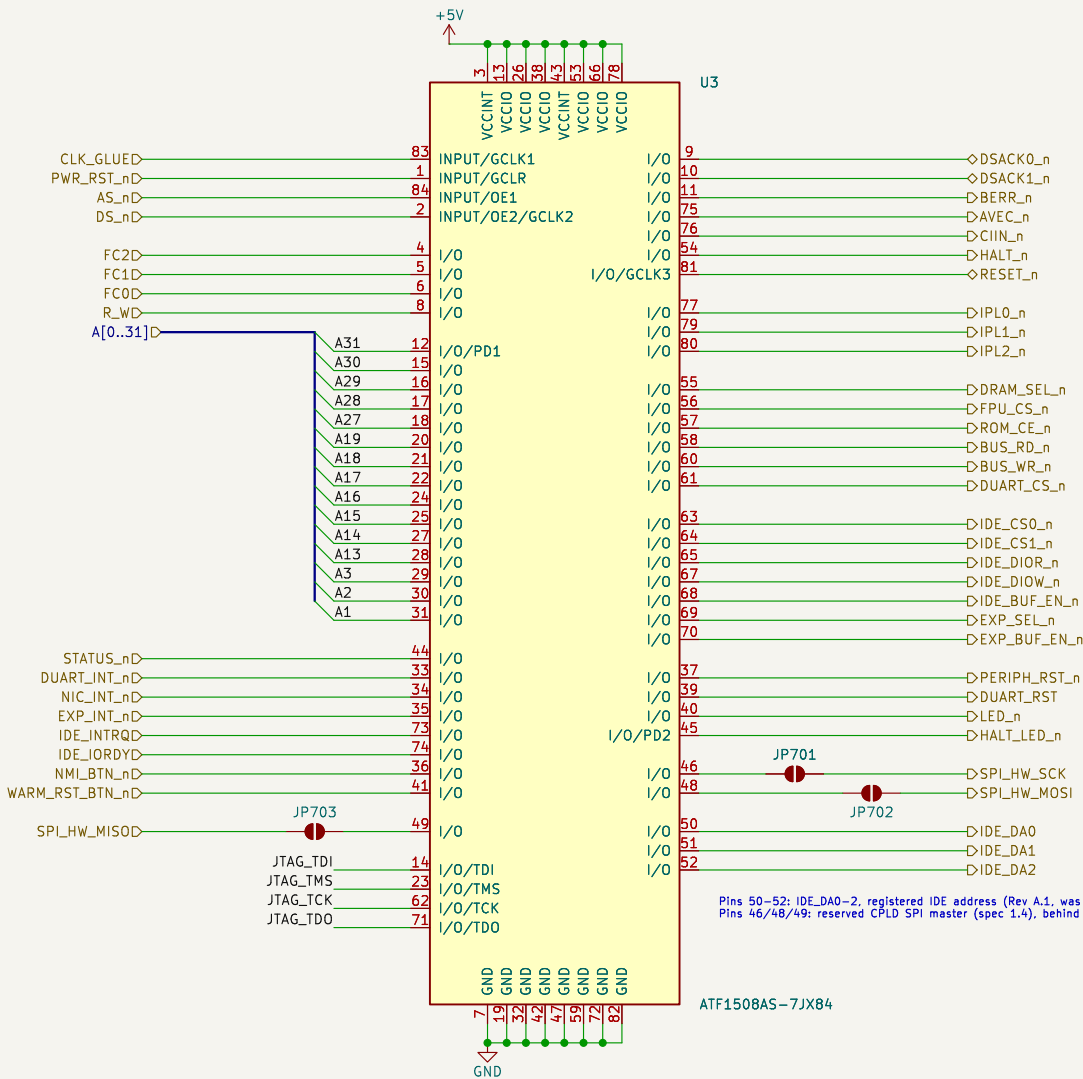
Date: 2026-09-26

Rev: A (draft)

Id: 6/13

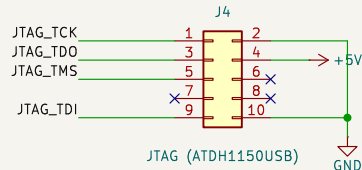
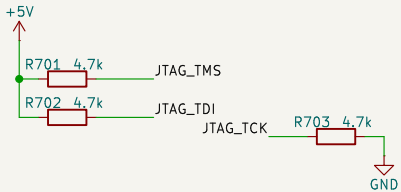
U3 glue / system controller (spec 2, 3, 4.2, 7.2)

FITTED PINOUT: Yosys 0.52 + hoglet67/attf150x.yosys -> Microchip fit1508.ave v1918 (Wine 10).
ATF1508AS PLCC84 -7, JTAG ON, pins locked (~preassign keep), -pin_keep on, -optimize off.
Rev A.1 (2026-09-27): 121/128 macrocells, 62 flip-flops, 322 product terms: 57/60 user I/O + 4 JTAG + 4/4 dedicated inputs.
Sources, reports, JEDEC: cpld/glue/ (glue.v, glue.ftt, glue.jed, pinout.csv).
No series resistors on U3 outputs (spec calls for none; add after scope checks if needed).
Open-drain nets (RESET_n, HALT_n) are emulated with OE: DSACKx/AVEC/BERR are driven low, driven high when AS_n rises and released <= 20.5 ns later (async clear; pull-ups on CPU sheet).
FPU_CS_n / EXP_SEL_n / EXP_BUF_EN_n: one product term incl. AS_n (no stale-decode runt, spec 3.2).

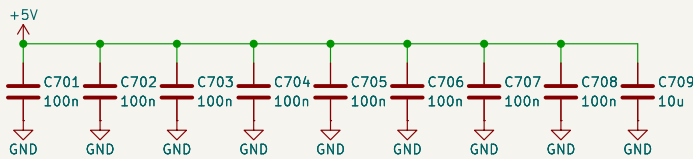


JTAG ISP header

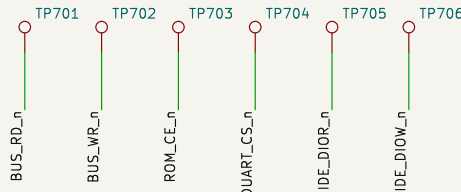
JTAG J4 (U3). Pinout: Microchip ATDH1150USB user guide (Atmel-B909A) Table 1 / Fig. 4.
VCC1 = +5V (the ATF1508AS VCCIO). TMS/TDI also have the device's internal pull-up option enabled in the fit (-tdi_pullup/-tms_pullup on); R701/R702 4.7k pull-ups and R703 4.7k TCK pull-down keep the TAP idle with no cable / blank part [EST values].



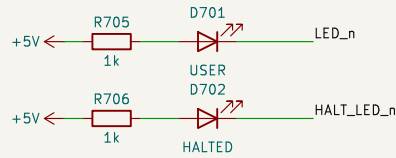
U3 decoupling: 8x 100 nF, one at each VCC pin (VCCINT 3, 43; VCCIO 13, 26, 38, 53, 66, 78), + 10 uF bulk



Test points (key strobes)



Status LEDs (U3 sinks: ~3 mA)



Interrupt input pull-up (spec 3.5)

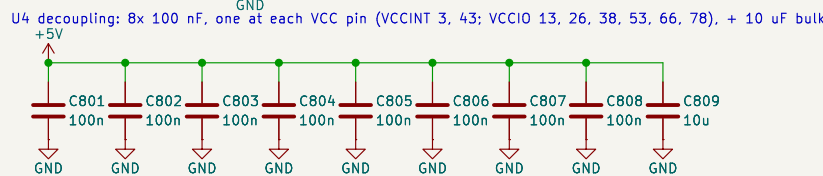
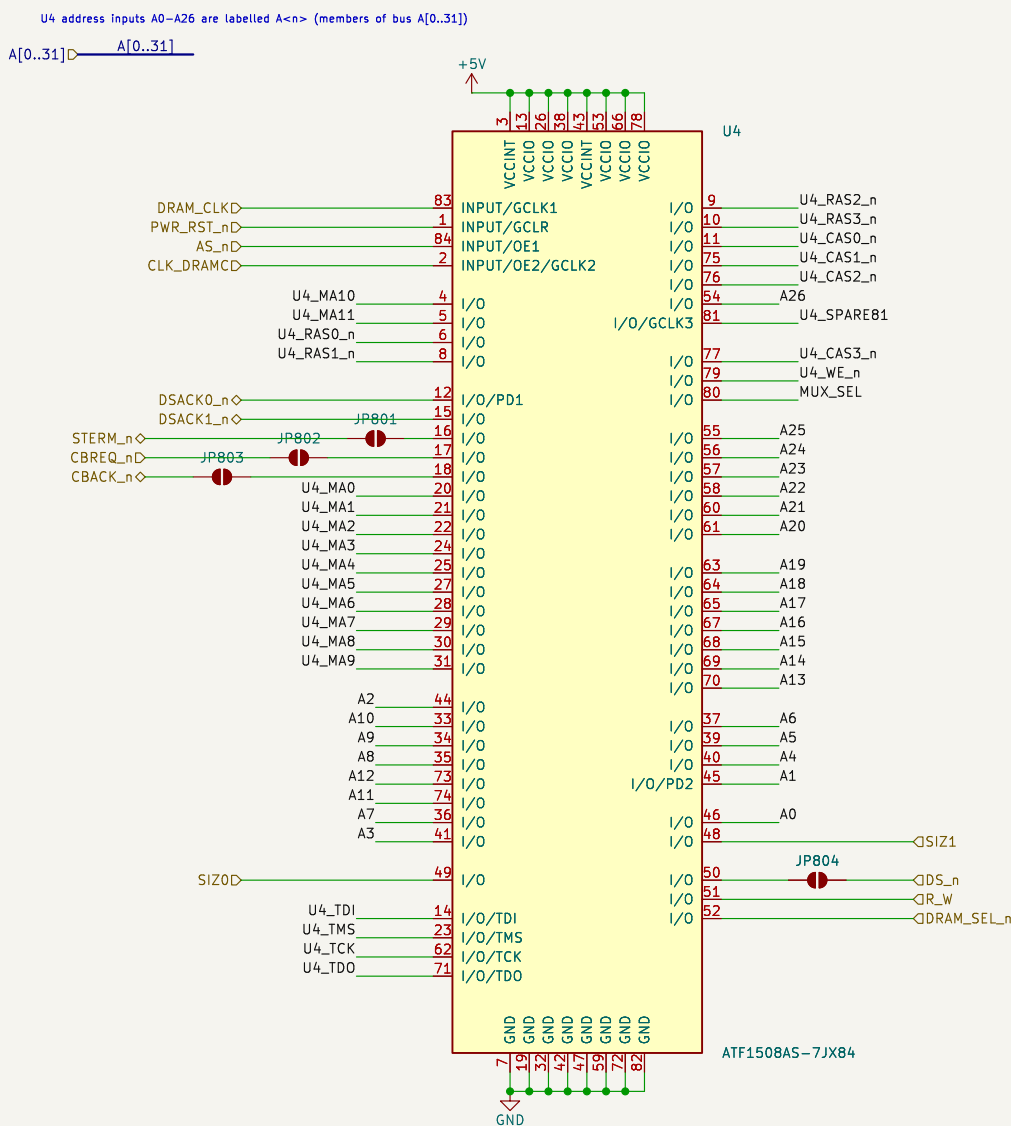
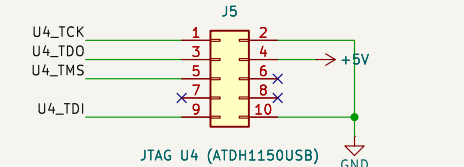
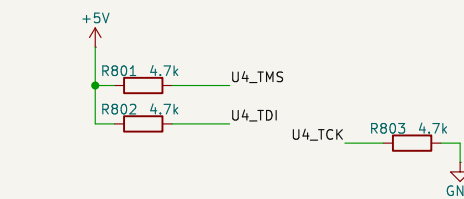


Logic: cpld/glue/glue.v; build: cpld/glue/build.sh
PLCC-84 pins vs doc0784 Rev.0784P; J4 per ATDH1150USB UG
FITTED pinout: fit1508 v1918, cpld/glue/pinout.csv (Rev A.1: 121/128 MC)
Matthew Bettcher - 68030 Linux SBC
Sheet: /glue_cpld/
File: glue_cpld.kicad_sch
Title: Glue/system CPLD U3: ATF1508AS-7JX84 (PLCC-84)
Size: A3 Date: 2026-09-26 Rev: A (draft)
KiCad E.D.A. 9.0.2 Id: 7/13

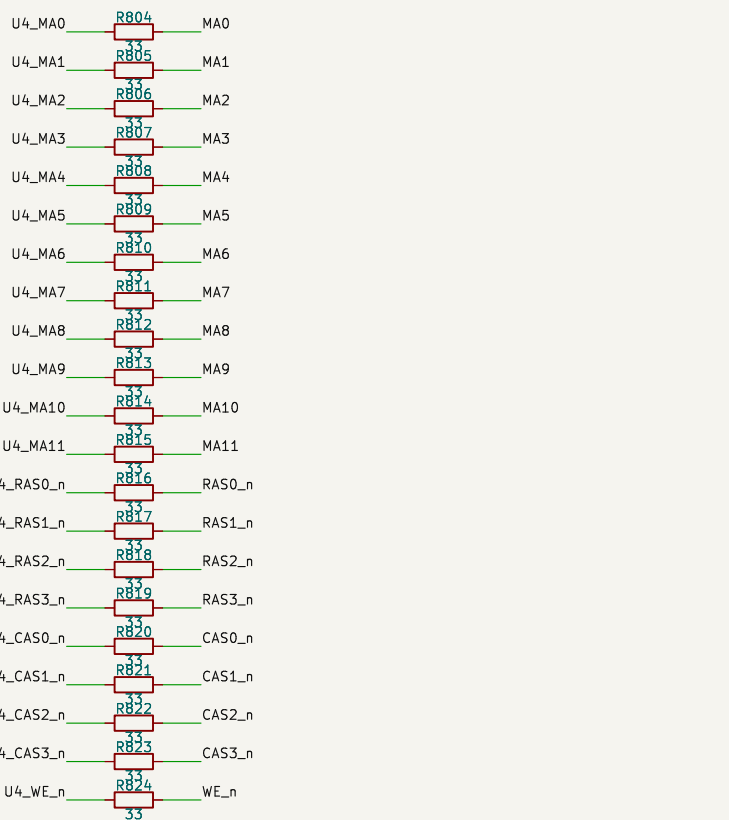
FITTED PINQOUT: `yours + hogle/!atf15x.yoys --> Microchip!1f1508.eve v1918 (wine), ATF1508AS PLCC84 ~7, JTAG ON, pins locked ~(pre-guess keep) ~optimize on ~pin_keep on. Result: 80/128 macrometers, 52 flip-flops, 210 product terms; 55/50 fanout /O + 4 JTAG + 5/4 dedicated inputs. Worst register-to-register 14 ns (50 MHz = 20 ns met).`
Sources, reports, JEDEC, testbench: `cpid/dram/ (dram.v, dram.fim, dram.jed, timing.tb, sim/).`
Logic after Mackerel~30 dram.controller. (C. Maykish, MIT): 50 MHz FSM, 2-flap AS synchronizer, CBR refresh every 7.5 us alternating sides. 100 us + 16 CBR power-up, CAS lanes from SIZ1/SIZ0/AM1_0, DSAK1+0 (32-bit port). Pins 16/17/18/50 (STERN_n/CBREQ_n/CBACK_n/DS_n) are unused in Rev A: reached only through open solder jumpers. Pins 2 (GCLK2) carries CLK_DRAMC for a future CPU-synchronous (burst) FSM; pin 81 spare ~> TP809.

Series resistors: 33R source termination on MA/RAS/CAS/WE. Trace ZO ~50-60R minus the CPLD output impedance (-15-25R) [EST]: SIMM input loads (Micron): MA 48/95 pF, WE 64/127 pF, RAS 32 pF, CAS 16/32 pF (1/2-sided) -> 33R x 100 pF = 3.3 ns edge slowing, inside the DRAM timing margins (TRAH/TASC/TAHA >= 14 ns post-fit).

J5: separate header for U4 (same pinout as J4 on the glue sheet: ATDH1150USB UG Table 1). Not chained with U3: the ATDH1150USB/ATMISIP support multi-device chains, but a separate header lets each CPLD be programmed/recovered alone and a blank or damaged part cannot break the other's chain. RB01/RB02 4.7k pull-ups, RB03 4.7k TCK pull-down [EST values].



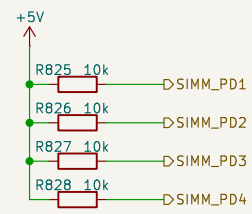
33R, placed next to U4 (source termination



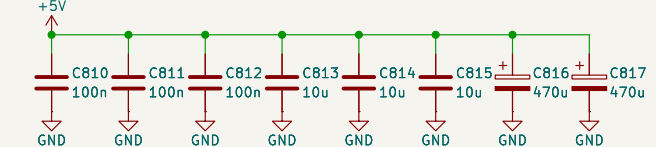
R806-R815 (MA2-MA11): remove if the 74ACT257 fallback (UB01-UB03) is fitted

The diagram shows eight signals over time, each with a red circle indicating a specific event. The signals are: RAS0_n, RAS1_n, CAS0_n, CAS1_n, CAS2_n, CAS3_n, WE_n, MUX_SEL, and J4_SPAREB1. The signals are labeled TP801 through TP805.

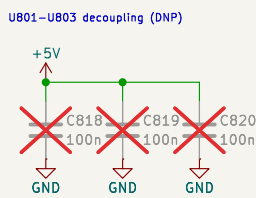
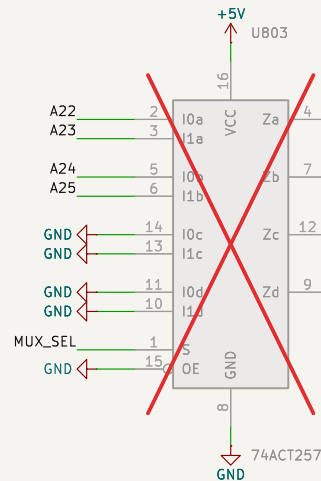
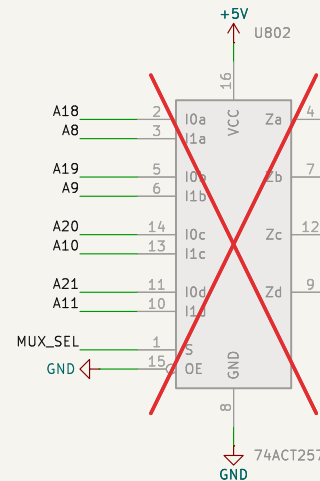
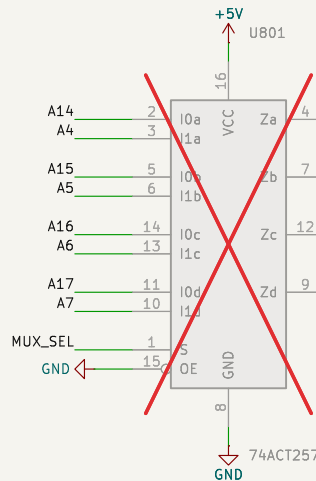
SIMM presence detect (spec 5.1): 10k pull-ups:
module ties PRDn to VSS or leaves it open.
Read on DUART IP3-IP6 (advisory; firmware sizes by probing)



SIMM decoupling (spec 6.3): 100 nF + 10 uF at each VDD pin (10, 30, 59), 2x 470 uF low-ESR at the socket ends for the CBR refresh current spikes (all 16/32 DRAMs refresh at once)



Not needed: U4 fits with the MAMmux Inside (80/128 MC, 50 MHz met). To use: fit U801-U803 + C818-C820 and R829-R838, remove R806-R815 (U4 MA2-MA11 path), and rebuild dram.v with MA2-MA11 unused. S = MUX_SEL (U4 pin 80; low = row on 10, high = column on 1). 74ACT257 tpd <= 10.5 ns (onsemi! MC74ACT257), vs 7.5-10 ns in U4

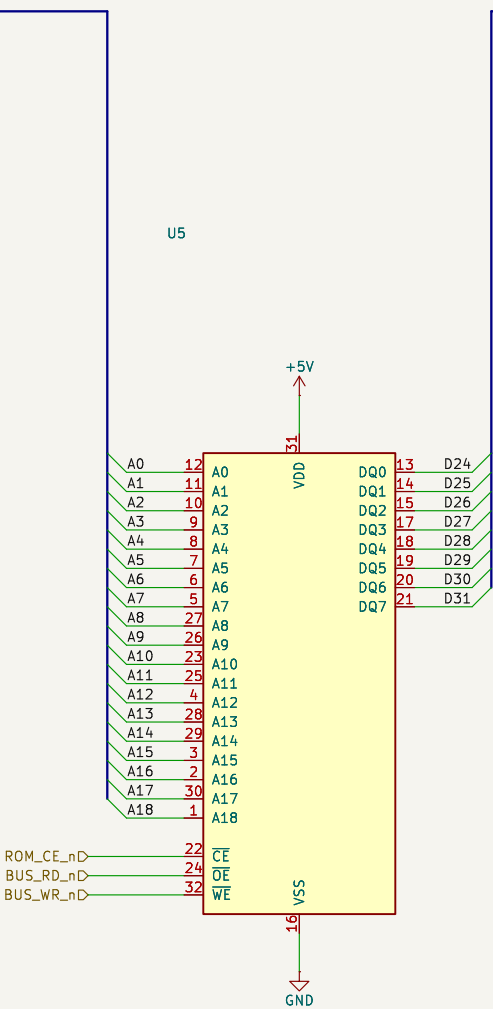


Logic: cpld/dram/dram/after Mackerel--30_dram_controller.v; build: cpld/dram/build.sh
 SIMM pinout: Micron MT80432/MT160B832; socket TE 582201--4 (hole pattern ENG-582201-Q1, B1)
 FITTED pinout: fit1508 v1918; cpld/dram/pinout.csv (80/128 MCM, 50 MHz met)
Pinout: Bettcher -- 68030 SBC
 Sheet: /dram/
 File: dram/kicad_sch
Title: DRAM controller CPLD U4 (ATF1508AS-7JX84) + 72-pin SIMM J1
 Size: A2 Date: 2026--09--26 Rev: A (draft)
 KiCad E.D.A. 9.0.2 Id: 8/13

SST39SF040 boot ROM (spec §2.2, §3.2, §8)

A0–A18 (A19–A31 are on the bus but not used by the 512 KB part; the glue alias is in the decode)

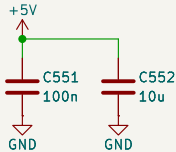
DQ0–DQ7 = CPU D24–D31 (8-bit part, DSACK0 only)



SST39SF040–70–4C–NHE

Orderable part SST39SF040–70–4C–NHE (PLCC–32, 70 ns, commercial). The –70–4C–PHE is the DIP–32 option and is NOT this pinout. Socket footprint is the PLCC–32 THT socket (BOM XU5); the chip sits in it. Read timing, 25 MHz, 1 wait state, no buffer on this bus (DS20005022C Table 11, 70 ns column): TCE = TAA = 70 ns, TOE = 35 ns, CE at the pin <= 26.5 ns (spec §3.2) so data <= 96.5 ns, needed by 118 ns; 21.5 ns margin, OE path: BUS_RD at 70 ns + TOE 35 + 1 ns board = 106 ns, margin 12 ns, TRC min 70 ns; a 1–WS cycle is far longer, 33 MHz still needs ROM_WS = 2 (spec §3.2). In–system program/erase: WE# is BUS_WR_n. The glue drives it only when FLASH_WE = 1 (spec §2.4). Byte program completes internally within 20 us (datasheet Byte–Program section); the CPU polls DQ7/DQ6.

Decoupling at the socket: 100 nF + 10 uF on the one VDD pin (PLCC pin 31)



WE# is BUS_WR_n, which the glue drives only while the FLASH_WE register is set 70 ns grade: Table 11 TCE/TAA 70 ns, TOE 35 ns, 1 WS at 25 MHz still holds (spec §3.2) PLCC pinout: Microchip DS20005022C Fig. 2 (SST39SF040 column), not the PDIP pinout		
Matthew Bettcher – 68030 Linux SBC		
Sheet: /rom/ File: rom.kicad_sch		
Title: Boot flash: SST39SF040–70 (PLCC–32) at E000_0000		
Size: A3	Date: 2026–09–26	Rev: A (draft)
KiCad E.D.A. 9.0.2		Id: 9/13

